

Introduction to Bioinformatics

Chris Miller, Ph.D.
Washington University in St Louis



Register Here!

<https://redcap.link/bfx26-27>

Bioinformatics Workshop 2025-2026

(aka bfx-workshop)



Applied Bioinformatics for Genomics I

(aka BIOL 5624)

Applied Bioinformatics for Genomics I

Course: BIOL 5624

1 Credit Hour DBBS course

- **50% grade:** Attendance
 - 75% (9 lectures) must be in person
 - 3 can be viewed via recordings
- **50% grade:** Assignments
 - choose 8 of the 10 assignments
 - due by the end of the second Friday after the lecture



Register for BFX

<https://redcap.link/bfx26-27>

icts-precisionhealth.wustl.edu

johnegarza@wustl.edu

Bioinformatics Workshop 2025-2026

Supported by – ICTS Precision Health

- We aim to catalyze genomic research by providing grant review, development services, guidance and resources for genomic researchers and genomics education in the community.

Cite the **NIH CTSA Grant #UL1 TR002345** when research is supported by ICTS/CTSA funding or any ICTS Core Services

BFX Workshop – contact John if you haven't received the following

- Slack access, welcome email, Outlook bfx-workshop-2025 group invite



Support Transdisciplinary Research

- Match clinicians and investigators
- Review grants
- Fund Precision Health Innovation awards



Develop Common Workflows

- Genomic consent and return of results
- Enable access to large genomic data sets
- Develop infrastructure to speed discovery translation



Educate the Community

- Expand access to educational research programs
- Educate scientific and clinical community in precision health
- Engage broader STL community in genomic research

Leadership Team



Megan Cooper,
MD, PhD



Precision Health Led Projects

 icts-precisionhealth.wustl.edu

 j.mckenzie@wustl.edu

- **Pilot funding & Research reviews**
 - Precision Health Innovation Awards; ICTS Research Development Program

- **Return of Results (ROR) for Research Participants**

- Genetic counseling, process for returning ACMG secondary results

- **Genomic Database Access and Submission**

- UK Biobank, All of Us Research Program, dbGaP, AnVIL, SRA
 - Assistance to submit human genomic data to shared repository

- **Institutional Genomic Consent**

- One Protocol One Consent, BJC-Webb electronic biobank

- **Community Education & Engagement**

- Precision Health for the Ages Workshop Series

Providing Support For

- **Core Services**

- WU Biological Therapy Core Facility (BTCF), McDonnell Genome Institute (MGI)

- **Informatics Tools for Precision Health**

- Bioinformatics Workshop (BFX), pVAC, CIViC,

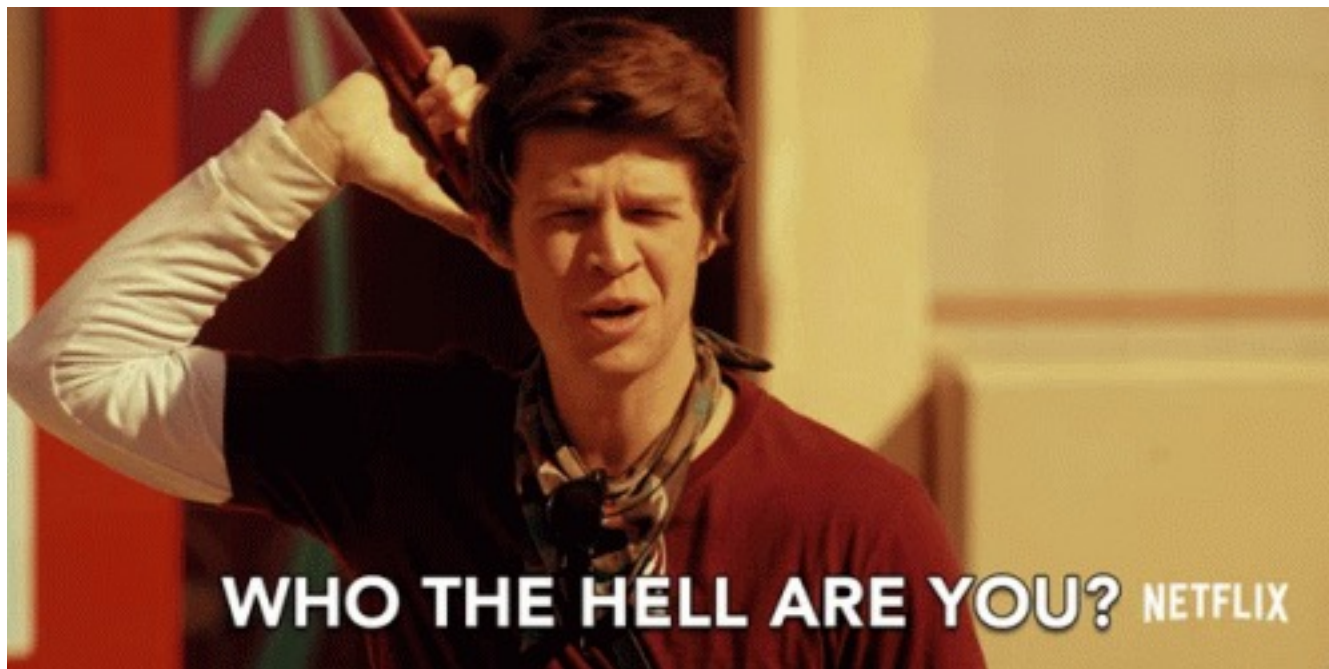
- **Communications and Outreach**

- Women in Innovation and Technologies (WIT) program, EQUALIZE program through OTM

- **Educational Opportunities**

- Precision medicine pathway, Bioinformatics Workshop (BFX), Genomics in Medicine





WHO THE HELL ARE YOU? NETFLIX

Who we are, and why you should trust us



Chris Miller, Ph.D.

Course Director

Associate Professor
Division of Oncology



John Garza

Course Coordinator/TA

Bioinformatics/Genome Analytics
Programmer

over 20 years of experience in
Bioinformatics and Computational Biology

Other Lecturers/Organizers include:

Jason Walker

Juan Macias

Obi Griffith

Brigida Rusconi

Jennifer Foltz

Jenny McKenzie – ICTS

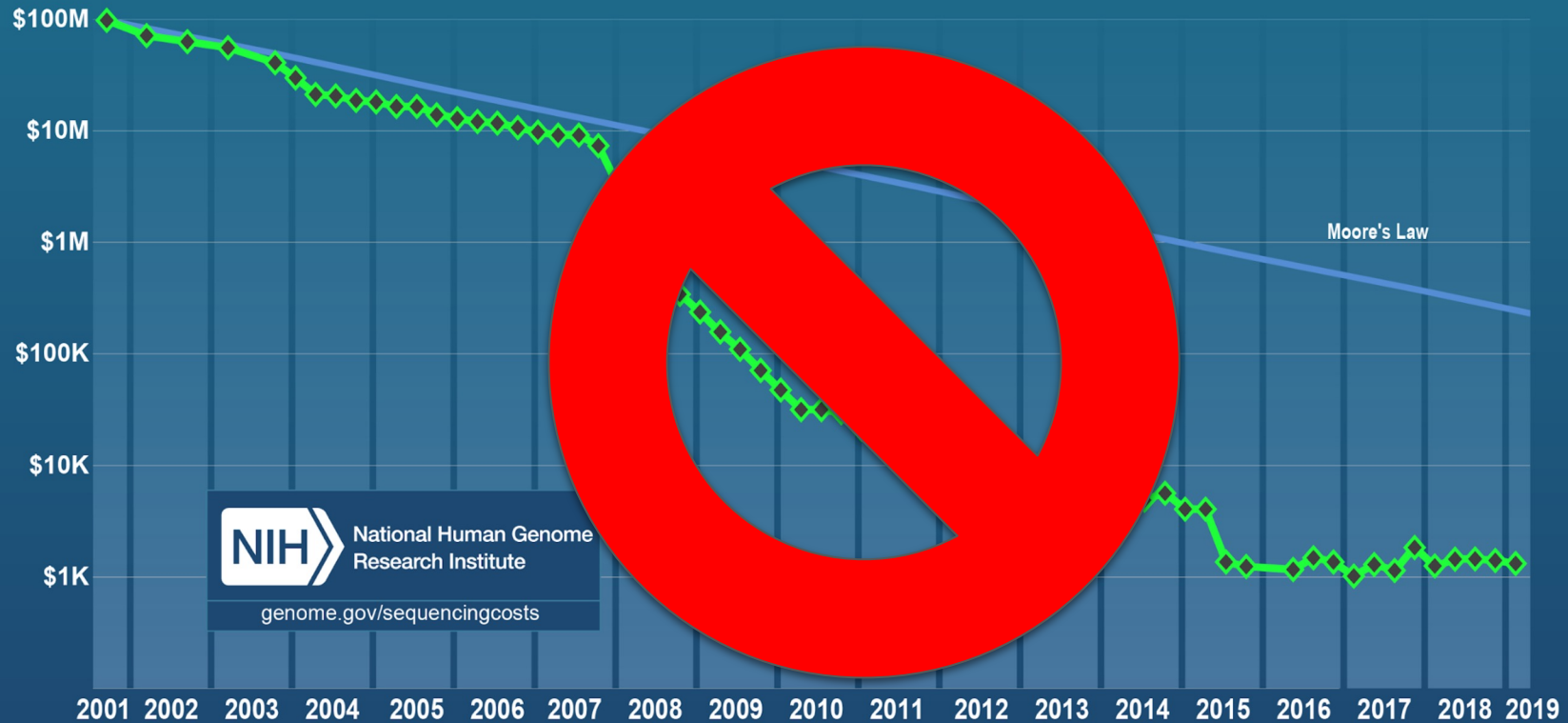
Precision Health Program Scientist

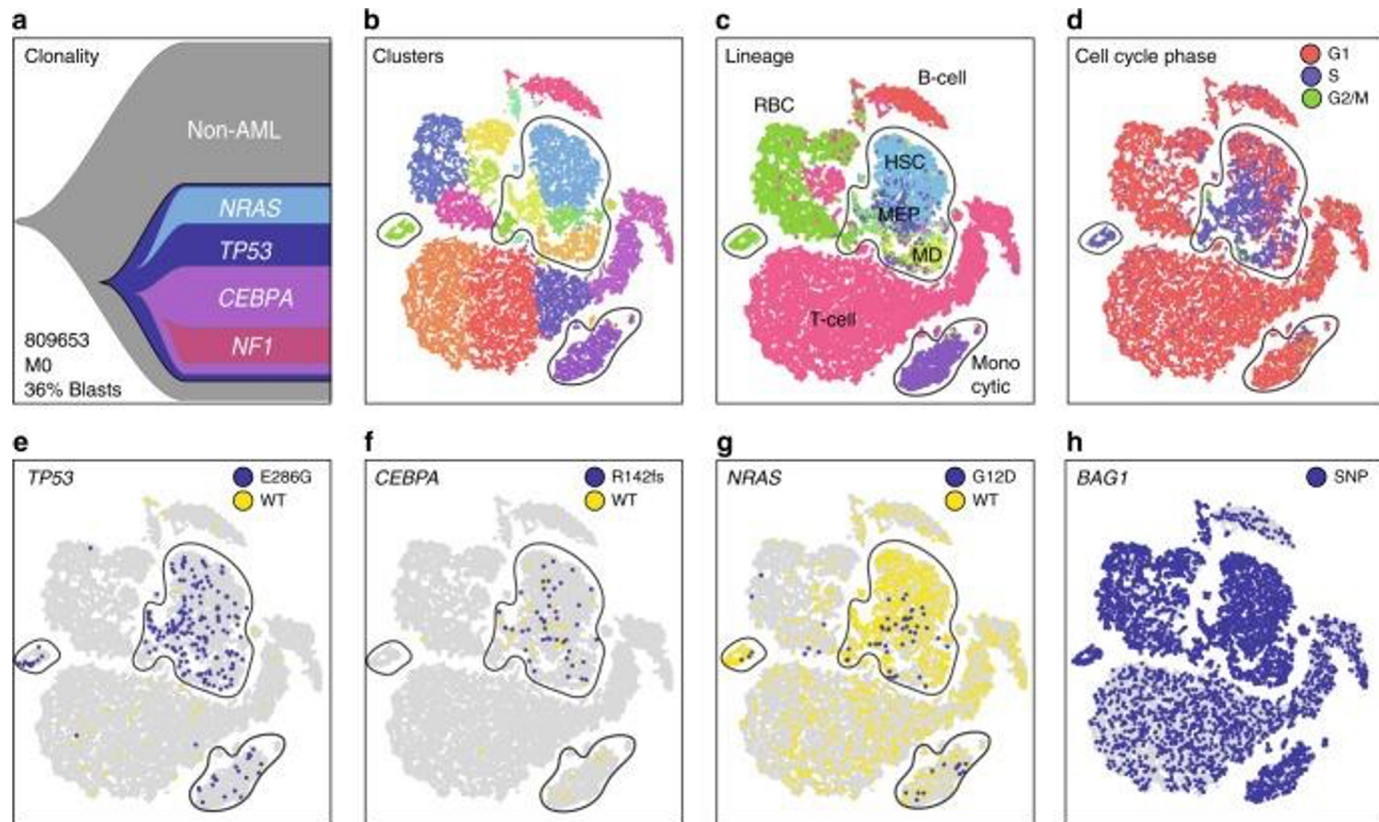
Why learn bioinformatics?

Why learn bioinformatics?

- Biology is now a quantitative discipline - especially genomics

Cost per Genome





Why learn bioinformatics?

- Biology is now a quantitative discipline - especially genomics
- Skills in programming, statistics, and visualization help you get the most out of your data



People who need complex data analysis

<https://hellogiggles.com/news/how-many-people-attend-coachella/>
<https://www.nytimes.com/2020/07/02/theater/germany-theater-coronavirus.html>



People who know how to do
complex data analysis

Why learn bioinformatics?

- Biology is now a quantitative discipline - especially genomics
- Skills in programming, statistics, and visualization help you get the most out of your data
- This course aims to teach you the theory and practice of computational biology, with a focus on genomics but lessons that apply broadly

What is bioinformatics?

What are the differences between bioinformatics and computational biology?



Bioinformatics vs. Computational Biology

[Computational biology](#), [bioinformatics](#) and [mathematical biology](#) are all interdisciplinary fields that combine biology with mathematics and computer science.

Computational Biology vs. Bioinformatics: What's the Difference?

By Samantha Costanzo Carleton

May 28, 2021

Exploring the Differences
Between Computational Biology
and Bioinformatics

Bioinformatics vs. Computational Biology



What is bioinformatics?

More Computational

More biological



Algorithm design

Building Pipelines

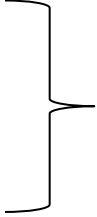
Developing Assays

Analysis of my
experiment

What is bioinformatics?

- Application of computational techniques to biological data
- Covers a lot of ground!
 - Population genetics
 - Cancer genomics
 - Microbial genomics
 - Proteomics
 - Ecology/Evolution
 - Medical informatics/EHR mining
 - computational behavioral biology
 - Epidemiology
 - Protein folding
 - CryoEM or tomography
 - Drug design/molecular dynamics
 - Algorithmic design/optimization
 - Metabolomics
 - Mathematical Biology

Common skills

- Statistics
 - Programming
 - Visualization
- 
- “Data science”
- Deep understanding of the biological system and experiments

Goals:

- To empower you to improve and expedite your research
- To expose you to new ideas and techniques that may advance your research program

Course structure

- Pair an introduction to a biological or technical concept with some of the tools needed to analyze it
- This week
 - Unix command line skills
- Next week:
 - Working on the RIS compute cluster
- Following week:
 - Sequence data generation
 - How to read, manipulate, and run quality control on sequence data
 - "how do I know if my experiment worked"

Homework

- Will not be turned in or graded unless you're taking the course for credit (though we will happily look at it with you)
- Will be useful for understanding subsequent lectures
- Posted on the course page

<https://github.com/genome/bfx-workshop>

"Office hours"

- 30 minutes before and after class each Monday
- Homework
- Concepts/experimental design
- Your own data/scripts

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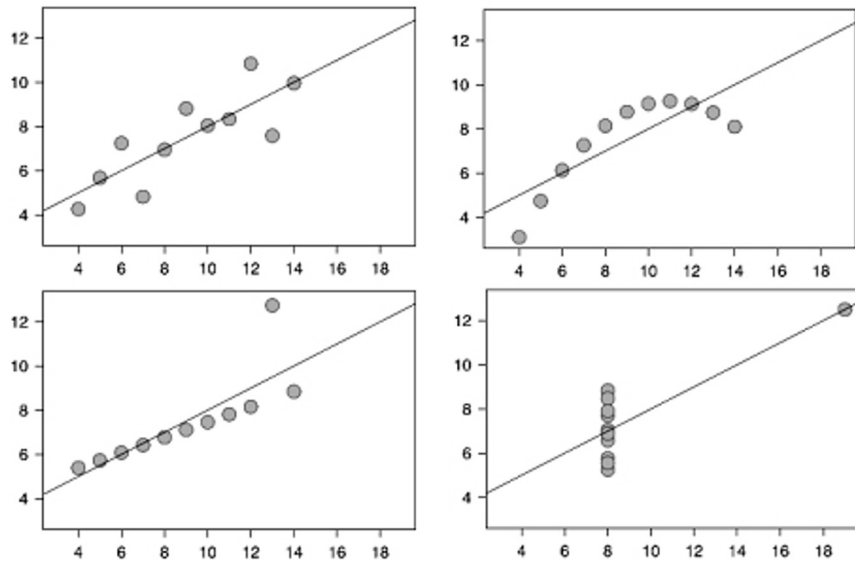


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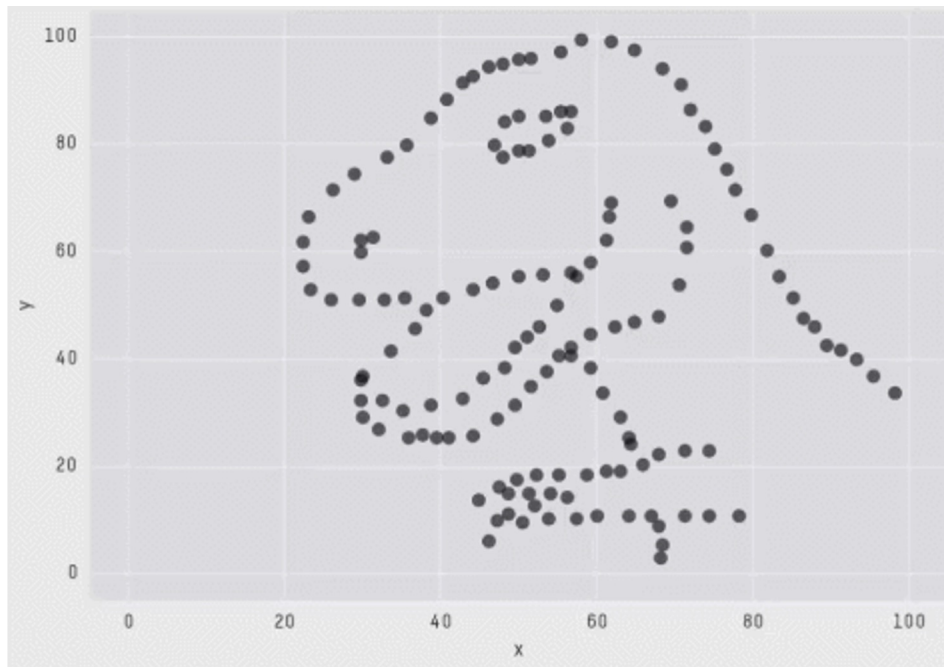
Don't trust your data

Trusting your data



Property	Value	Accuracy
Mean of x	9	exact
Sample variance of x : σ^2	11	exact
Mean of y	7.50	to 2 decimal places
Sample variance of y : σ^2	4.125	± 0.003
Correlation between x and y	0.816	to 3 decimal places
Linear regression line	$y = 3.00 + 0.500x$	to 2 and 3 decimal places, respectively
Coefficient of determination of the linear regression: R^2	0.67	to 2 decimal places

Datasaurus Dozen



X Mean: 54.2659224
Y Mean: 47.8313999
X SD : 16.7649829
Y SD : 26.9342120
Corr. : -0.0642526

Statistics can be dangerous

- Visualize your data!
- A picture is worth a thousand p-values

"If your experiment needs statistics, you ought to have done a better experiment"

- Ernest Rutherford

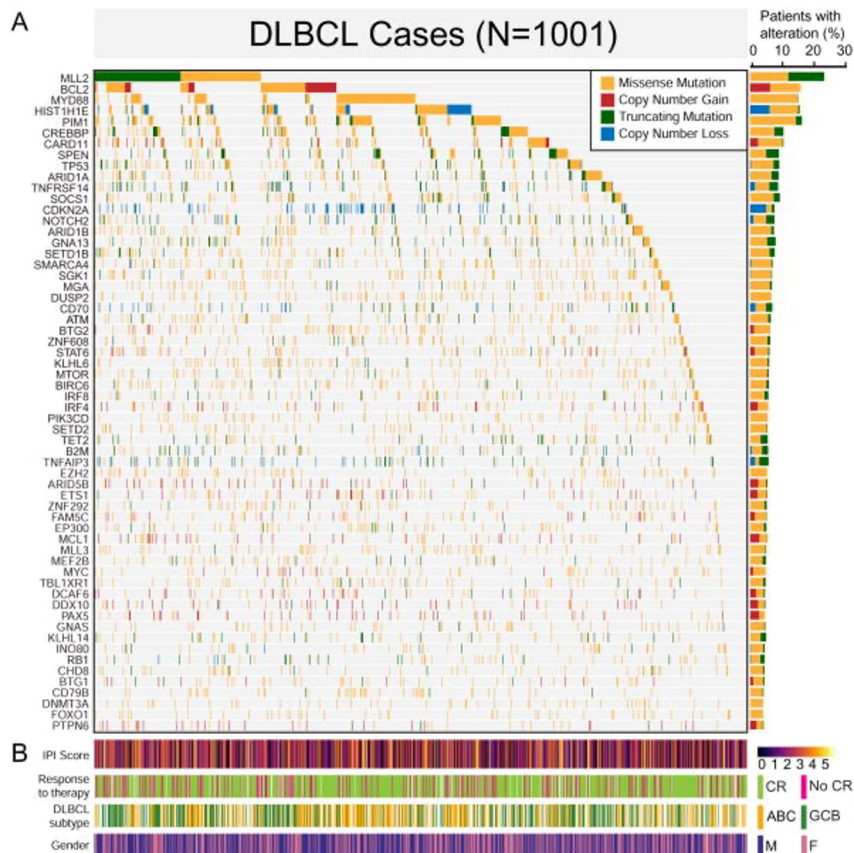
Can't quite co-sign, but he has a point:

If you can't make a plot convincing you that an effect is real, how confident are you, really?

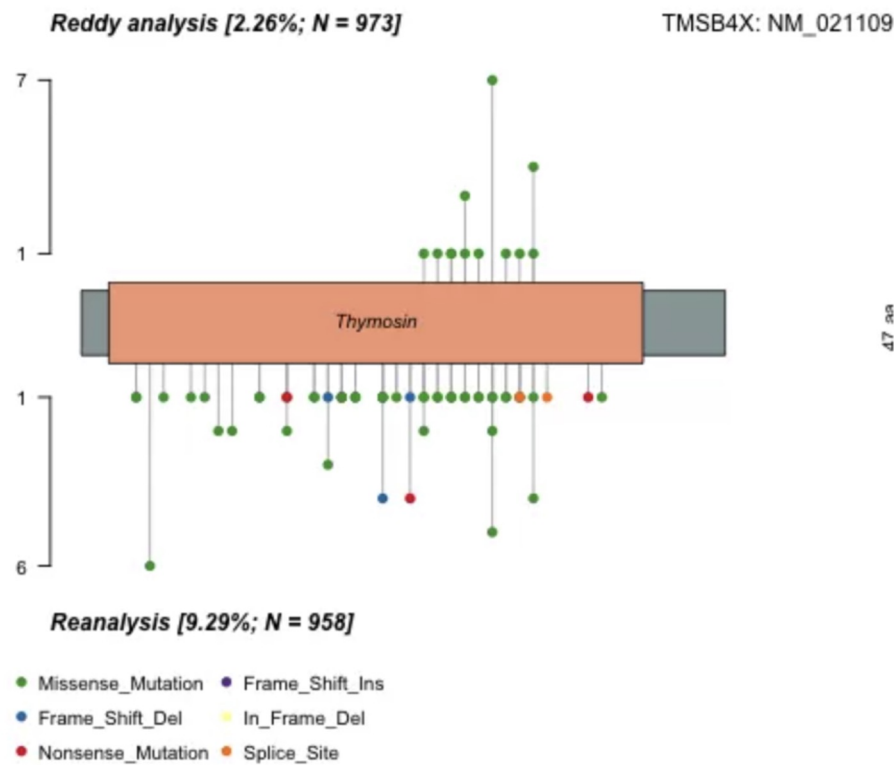
Dangerous situations:

- The bioinformatics core aligned the data and sent me a list of differentially expressed genes. I'm done, right?
- We ran Mutect to call somatic mutations in this tumor genome. Let's take it to the bank

Real world consequences



Real world consequences



Real world consequences

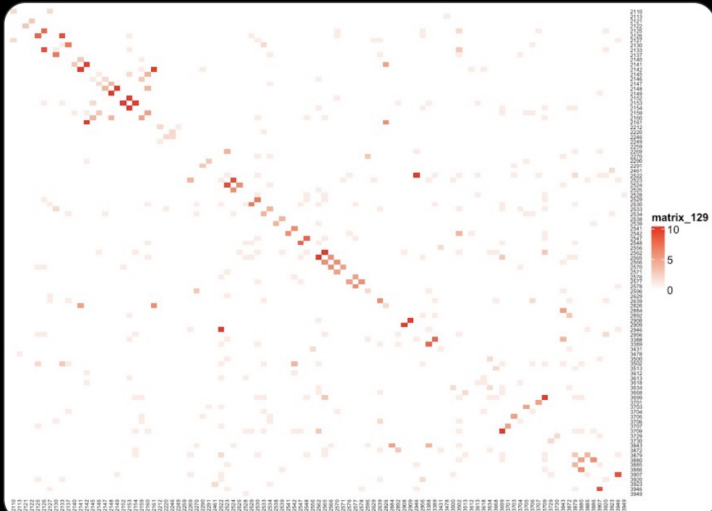
Ryan D Morin @morinryan · Oct 2

RNA/DNA mismatches (sample swaps) affecting at least 10% of the patients in Reddy et al, a Cell paper with over 700 citations. Same issue was described in a more recent paper from this group. [#lymphoma](#) [#genomics](#) [#goodresearchpractice](#) [pubpeer.com/publications/E...](#)

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Ryan D Morin @morinryan · Oct 2

Sharing of variants between RNA and DNA. Red should be on the diagonal. Most swaps seem to be between adjacent or nearby IDs.



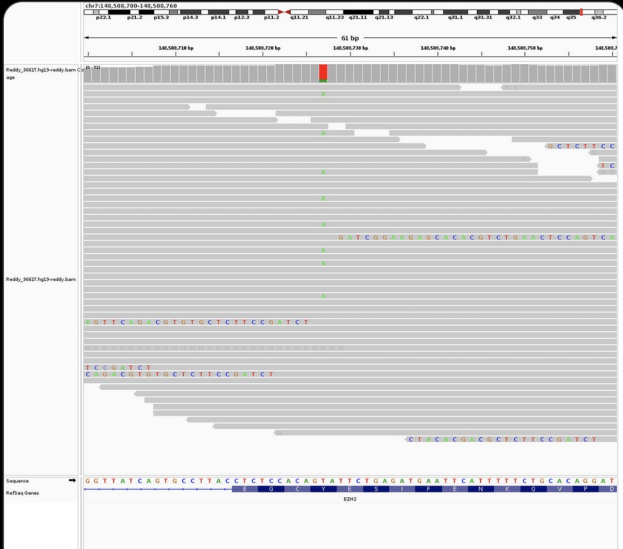
Ryan D Morin @morinryan · Nov 4

There are over 3,600 examples of variants like this, supported by at least 3 somatic variant callers (i.e. by consensus, they're real) and yet Reddy didn't report them. All of these are coding variants in the DLBCL genes described in Reddy but all were absent for some reason.

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Ryan D Morin @morinryan · Nov 4

24/33 (this is the limit imposed by Twitter). This is just the first 24. If someone still thinks I'm cherry-picking examples. This is a clinically relevant hot spot that was described 7 years before the Reddy study. Inexcusable to miss this many of them, and yet excuses are made!



Real world consequences

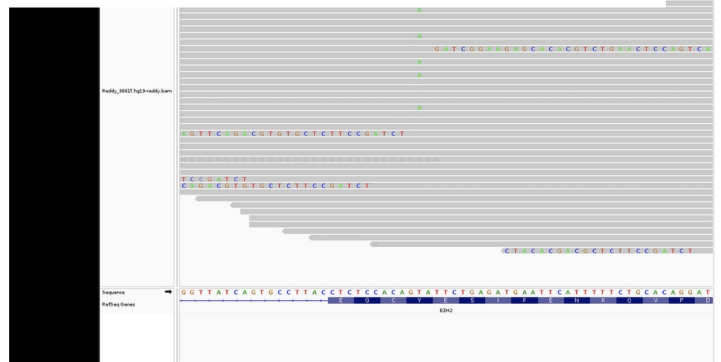
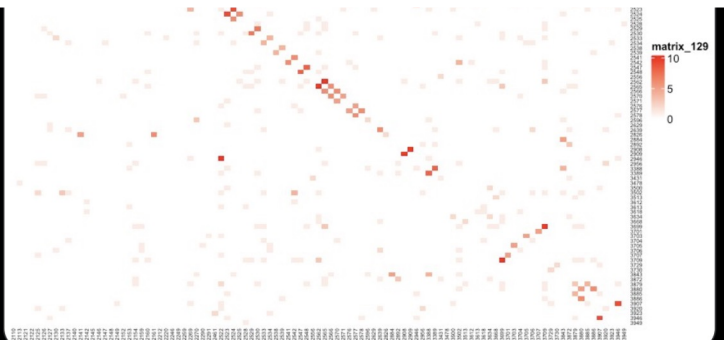
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Although the effects on each conclusion from Panea *et al* has not been evaluated, we demonstrated that ~30% of the reported mutations are not supported by their WGS data, which caused a significant inflation of the mutation prevalence of at least 16 genes and the rate of coding mutations in 9 genes (Supplemental Figure S3). These lead to



Errors

- Will happen!
- Errors of commission vs omission
- Type 1 errors – False positives
- Type 2 errors – False negatives

Lessons to be learned

- Check and double check and triple check your data and your scripts
- Bioinformatic experiments need controls too!
- Sanity checks
- Visualize your data!
- Admit when mistakes are made

“Analyzing your data means inherently distrusting your data until you have exhausted yourself into giving up and trusting it.”

-Aaron Quinlan

Artificial Intelligence/LLMs in biology

- AI is not machine learning
 - SVMs, random forest, etc - statistical techniques that have been around for a long time
- Claude, ChatGPT, Gemini, Grok, et al.

AI tools are a mixed bag for research

- Can be powerful assistants
 - Code generation
 - Proofreading
 - Mining the literature
 - Personalized learning

Can be a "cognitive crutch"

- AI often produces code/conclusions/data that are **confidently wrong**
- Science/medicine rely on being verifiably **correct**
- We can't spot errors we haven't learned to recognize

Expertise before Augmentation

- Run the tool, write the code, master the concept *first*
- Ask claude what the right statistical test is, but then consult a statistics course page to verify it
- Have codex write code for you, but then step through line by line and add comments, documenting that you understand each line
- **Get proficient at the skills first, then use AI to give you a boost**

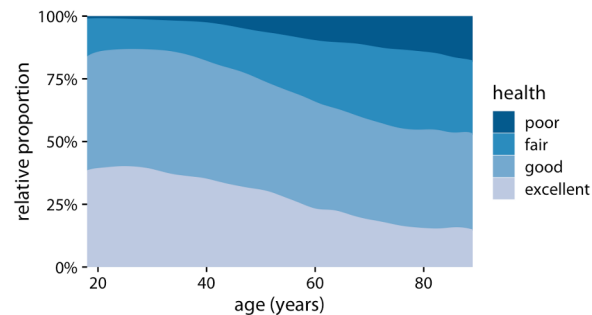
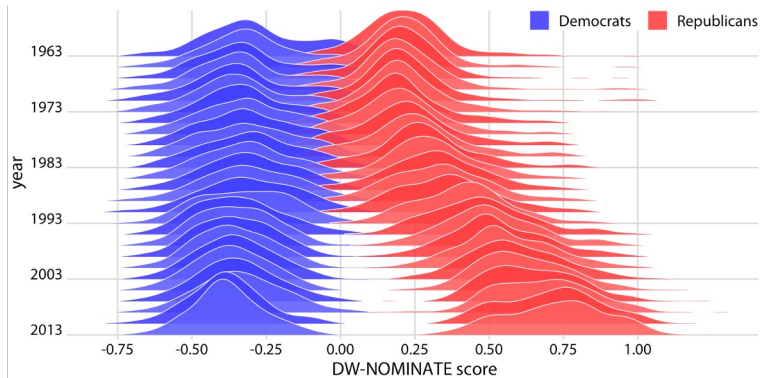
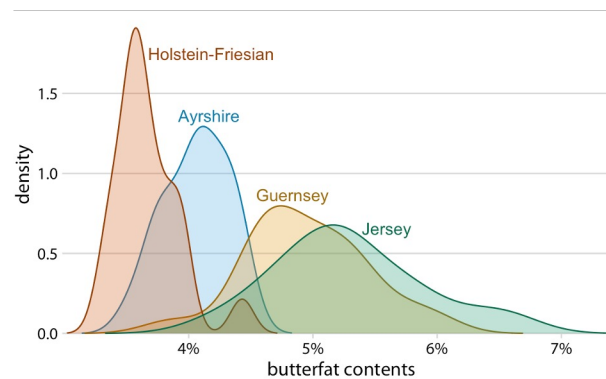
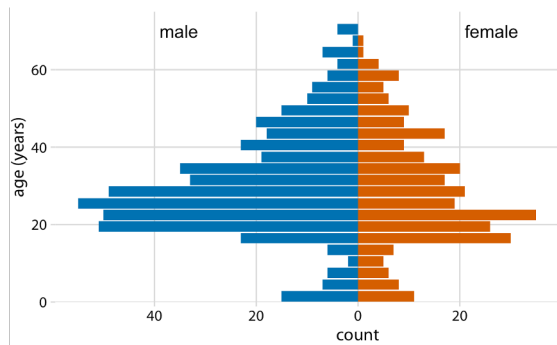
Accountable use of AI

- You are ultimately responsible for the analyses you publish

Useful ways to use AI in this course

- Orient you in unfamiliar territory
 - explain concepts (always ask for sources and verify key facts!)
 - summarize the literature (watch out for hallucinations!)
- Accelerate code generation/debugging
- Help brainstorm ideas – how can I transfer this concept to my work?

Turning your data into insight



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